(a) Start from your discipline

Geomorphology	**Hydrology**	Forest ecology	Biogeochemistry	Natural hazards
Where is fresh rock?	How much water is stored?	How much water can roots reach?	How thick is the reactive zone?	Is the regolith continuous?

Methods in the catalog: Electrical resistivity · Seismic refraction · Ground-penetrating radar · Electromagnetics · Borehole

(b) **Agent trace** Illustrative replay with sample values · nothing is computed — Replay

Request: "How much water is stored in the upper 5 m at this montane catchment site, and how did it change between the spring and late-summer surveys?"

- Understand the request — Mid-tier · 0.9 s
  Site: **a montane catchment site (sample)** · product **subsurface water storage** · electrical resistivity and electromagnetics.
- **Ground the request** — 7 passages · 3 sources
  Library docs and docstrings 2,604 · Curated catalog metadata 512 · Published studies 296 — Search over 3,412 chunks; the top 7 are kept.
  "Convert resistivity to water content using Waxman-Smits model." `PyHydroGeophysX/petrophysics/resistivity_models.py`
- Load skills — ERT · EM · Time-lapse · Petrophysics · Uncertainty — Procedures the agent reads when a task calls for them — 5 of 5 needed
- **Plan with structured output** — Mid-tier · 1.4 s
  Constrained decoding against plan schema v1.2; the plan validated on the first pass. 6 tool calls, 2 inferred values, each cited.
- Call registered tools — 6 calls
  `load_ert_surveys` → `invert_time_lapse` → `load_tdem_data` → `invert_tdem` → `convert_water_content` → `write_report`
- **Execute deterministically** — Pinned environment
  The validated plan runs in a pinned environment with no model in the loop, and writes the report, figures and data files with a provenance record.
- Deliver — **Sample values** — Open the result in the workspace →
  Storage, 0 to 5 m: **1,056 ± 38** mm of water, line mean — Seasonal change: **−110 ± 13** mm, between the spring and late-summer surveys
  ✓ Curator review required — After curator approval, deposit to HydroShare under CC BY 4.0, inherited from the source surveys.

Design preview. Values are illustrative and no computation is performed.